Calculus 2

Section 3

23.

$$\int \frac{\cos 2x}{\cos x + \sin x}\, dx$$

Use identity $\cos 2x = \cos^2 x - \sin^2 x = (\cos x - \sin x)(\cos x + \sin x)$:

$$\frac{\cos 2x}{\cos x + \sin x} = \cos x - \sin x$$

Thus:

$$\int (\cos x - \sin x)\, dx = \sin x + \cos x + C$$

$$\boxed{\sin x + \cos x + C}$$

24.

$$\int (\sin x + \cos x)^2\, dx$$

Expand:

$$\sin^2 x + 2\sin x \cos x + \cos^2 x = 1 + \sin 2x$$

Integrate:

$$\int 1\, dx + \int \sin 2x\, dx = x - \frac{1}{2}\cos 2x + C$$

$$\boxed{x - \frac{1}{2}\cos 2x + C}$$

25.

$$\int (x+1)^3 (x-1)^3\, dx$$

Combine:

$$[(x+1)(x-1)]^3 = (x^2-1)^3$$

Expand:

$$(x^2-1)^3 = x^6 - 3x^4 + 3x^2 - 1$$

Integrate:

$$\int x^6 dx = \frac{x^7}{7}, \quad \int -3x^4 dx = -\frac{3x^5}{5}, \quad \int 3x^2 dx = x^3, \quad \int -1 dx = -x$$

$$\boxed{\frac{x^7}{7} - \frac{3x^5}{5} + x^3 - x + C}$$

$$\int \left(x^2 - \frac{1}{x}\right)^6 x^8 \, dx$$

Factor x^8:

$$\int \left(x^2 - \frac{1}{x}\right)^6 x^8 dx = \int \left(x^2 - x^{-1}\right)^6 x^8 dx$$

Better: factor x^8 inside the 6th power?
Better: Let $u = x^2 - \frac{1}{x}$, but too messy. Let's try factoring x^2:

Inside: $x^2 - \frac{1}{x} = \frac{x^3-1}{x}$. Then:

$$\left(\frac{x^3-1}{x}\right)^6 x^8 = (x^3-1)^6 x^{8-6} = (x^3-1)^6 x^2$$

Now substitute $u = x^3 - 1$, $du = 3x^2 dx \Rightarrow x^2 dx = \frac{du}{3}$:

$$\int u^6 \cdot \frac{du}{3} = \frac{1}{3} \cdot \frac{u^7}{7} + C = \frac{u^7}{21} + C$$

Back-substitute:

$$\boxed{\frac{(x^3-1)^7}{21} + C}$$

27.

$$\int \frac{dx}{4 + x^2}$$

Standard form: $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

Here $a^2 = 4 \Rightarrow a = 2$:

$$\boxed{\frac{1}{2} \tan^{-1} \frac{x}{2} + C}$$

28.

$$\int \frac{dx}{1 + 9x^2}$$

Write as $1 + (3x)^2$, let $u = 3x$, $du = 3dx \Rightarrow dx = \frac{du}{3}$:

$$\int \frac{du/3}{1+u^2} = \frac{1}{3} \tan^{-1} u + C$$

Back-substitute:

$$\boxed{\frac{1}{3} \tan^{-1}(3x) + C}$$

29.

$$\int \frac{dx}{\sqrt{16 - 9x^2}}$$

Factor: $\sqrt{16 - 9x^2} = \sqrt{16\left(1 - \frac{9x^2}{16}\right)} = 4\sqrt{1 - \left(\frac{3x}{4}\right)^2}$

Let $u = \frac{3x}{4}$, $du = \frac{3}{4}dx \Rightarrow dx = \frac{4}{3}du$:

$$\int \frac{4\,du/3}{4\sqrt{1-u^2}} = \frac{1}{3} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{3} \sin^{-1} u + C$$

Back-substitute:

$$\boxed{\frac{1}{3} \sin^{-1}\left(\frac{3x}{4}\right) + C}$$

30.

$$\int \frac{e^x\,dx}{\sqrt{1-e^x}}$$

Let $u = 1 - e^x$, $du = -e^x dx \Rightarrow e^x dx = -du$:

$$\int \frac{-du}{\sqrt{u}} = -\int u^{-1/2} du = -2\sqrt{u} + C$$

Back-substitute:

$$\boxed{-2\sqrt{1-e^x} + C}$$

31.

$$\int \frac{\cos x}{a^2 + \sin^2 x}\,dx$$

Let $u = \sin x$, $du = \cos x dx$:

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\frac{u}{a} + C$$

Back-substitute:

$$\boxed{\frac{1}{a}\tan^{-1}\left(\frac{\sin x}{a}\right) + C}$$

32.

$$\int \frac{dx}{x\sqrt{4 - 9\ln^2 x}}$$

Let $u = \ln x$, $du = \frac{1}{x}dx$:

$$\int \frac{du}{\sqrt{4-9u^2}} = \int \frac{du}{\sqrt{4\left(1 - \frac{9u^2}{4}\right)}} = \frac{1}{2}\int \frac{du}{\sqrt{1 - \left(\frac{3u}{2}\right)^2}}$$

Now let $v = \frac{3u}{2}$, $dv = \frac{3}{2}du \Rightarrow du = \frac{2}{3}dv$:

$$\frac{1}{2}\int \frac{(2/3)dv}{\sqrt{1-v^2}} = \frac{1}{3}\int \frac{dv}{\sqrt{1-v^2}} = \frac{1}{3}\sin^{-1} v + C$$

Back-substitute $v = \frac{3u}{2} = \frac{3\ln x}{2}$:

$$\boxed{\frac{1}{3}\sin^{-1}\left(\frac{3\ln x}{2}\right) + C}$$

$$\int \frac{x - \sqrt{\sin^{-1} x}}{\sqrt{1-x^2}}\,dx$$

Split:

$$\int \frac{x}{\sqrt{1-x^2}}\,dx - \int \frac{\sqrt{\sin^{-1} x}}{\sqrt{1-x^2}}\,dx$$

First part: Let $u = 1 - x^2$, $du = -2x dx \Rightarrow x dx = -\frac{du}{2}$:

$$\int \frac{x}{\sqrt{1-x^2}}dx = \int \frac{-du/2}{\sqrt{u}} = -\frac{1}{2}\cdot 2\sqrt{u} = -\sqrt{1-x^2}$$

Second part: Let $v = \sin^{-1} x$, $dv = \frac{dx}{\sqrt{1-x^2}}$:

$$\int \sqrt{v}\,dv = \frac{2}{3}v^{3/2} = \frac{2}{3}(\sin^{-1} x)^{3/2}$$

So overall:

$$-\sqrt{1-x^2} - \frac{2}{3}(\sin^{-1} x)^{3/2} + C$$

$$\boxed{-\sqrt{1-x^2} - \frac{2}{3}(\sin^{-1} x)^{3/2} + C}$$